What did Snowball actually learn?

A paired rollout audit of validation at steps 0 and 384

16 October 2026 · 100 questions · 200 archived responses · 14 task categories

The improvement is mixed. There is concrete evidence of better coordinate retrieval, distance arithmetic and some salt-bridge calculations. But several higher-scoring categories are explained by class defaults or an easier geometric proxy. The rotamer responses now attempt dihedral calculations, yet their mathematical correctness is much weaker than their final-bin reward suggests.

The strongest shortcut results: step 384 says solvent-exposed in all 118 parseable S03 responses and coil in all 110 parseable S04 responses. In G03, 104 of 108 answers agree with the nearest-alpha-carbon proxy, although the task asks for minimum heavy-atom distance.

Rotamers: all 13 reviewed step-384 traces attempt numerical geometry; none is verified correct end to end. Seven still earn reward 1. A recurring missing factor changes angles but leaves all 116 validation gold bins unchanged when applied to the correct coordinates. This particular error is invisible to the category-only reward on these tasks.

This supplement qualifies the molecular-reasoning interpretation of the original result report; it does not change its measured scores. The matched original-benchmark result remains 5→11 of 40 strictly, or 8→11 after the existing box correction, with wide uncertainty. The present archival validation audit uses different inference paths at the two checkpoints and must not replace that matched comparison.

The accompanying HTML viewer contains every full prompt and response beside its review. The compressed JSON contains the same frozen sample for machine inspection. No new model inference or GPU job was required for this audit.

Sampling, scope and interpretation

I selected **100 question IDs before inspecting their reasoning**, then reviewed both archived answers: 100 step-0 rollouts and 100 step-384 rollouts. Pairing the questions lets us compare how each answer changes. Python random.Random uses seed **20261016**; the manifest was frozen at 2026-10-16T17:50:18.558924+00:00.

Sampling is random within strata but deliberately disproportionate. Each of the 14 present families receives seven questions, except G04, which has only three and is reviewed in full, and S09, which receives 13 because the user specifically requested rotamer analysis. Within a family, quota is allocated as evenly as possible across the available joint success states, subject to each stratum's capacity; ties use a seeded shuffle. IDs are sampled uniformly without replacement inside each stratum.

The strata are **00** (both fail), **01** (step-0 fail, step-384 succeed), **10** (step-0 succeed, step-384 fail), and **11** (both succeed). For example, S09 contributes three 00, three 01, three 10 and four 11 questions. Its sample therefore scores 7/13 at each checkpoint by construction, even though full-category accuracy increases. G02 includes all four step-384 successes. **Do not estimate population accuracy from this sample.** All reported population scores and controls use the full category.

The underlying validation cohort is the same frozen 1,003-question, 20-source-group set; the sample spans 17 groups. Questions sharing a structure are correlated. I01 (interface contacts), S06 (ligand contacts), S07 (metal coordination), S08 (disulfide partner) and T01 (two-state comparison) have no eligible validation questions and cannot be assessed here.

Step 0 means the original Snowball checkpoint before this project's additional RL, not an untrained model. Step 384 is the validation-selected checkpoint requested by the user. The original-model answers come from baseline-validation-v2 standalone serving; step-384 answers come from the native training-evaluator archive. Both use the same frozen prompts, temperature 0, one answer and the remaining 32,768-token context. Their serving/execution paths differ. Thus observed differences are descriptive evidence, not a clean causal decomposition into training versus runtime effects.

What was checked

The sampler checks question membership, exact baseline request messages, baseline output budgets and native prompt hashes. Replaying the frozen native verifier reproduces all 2,006 recorded rewards. It saves the 100 selected IDs, all response SHA256 hashes, input/native-archive hashes and full prompts. Strict success includes the native requirements for formatting, truncation and refusal; a semantically correct but unparseable answer can score zero.

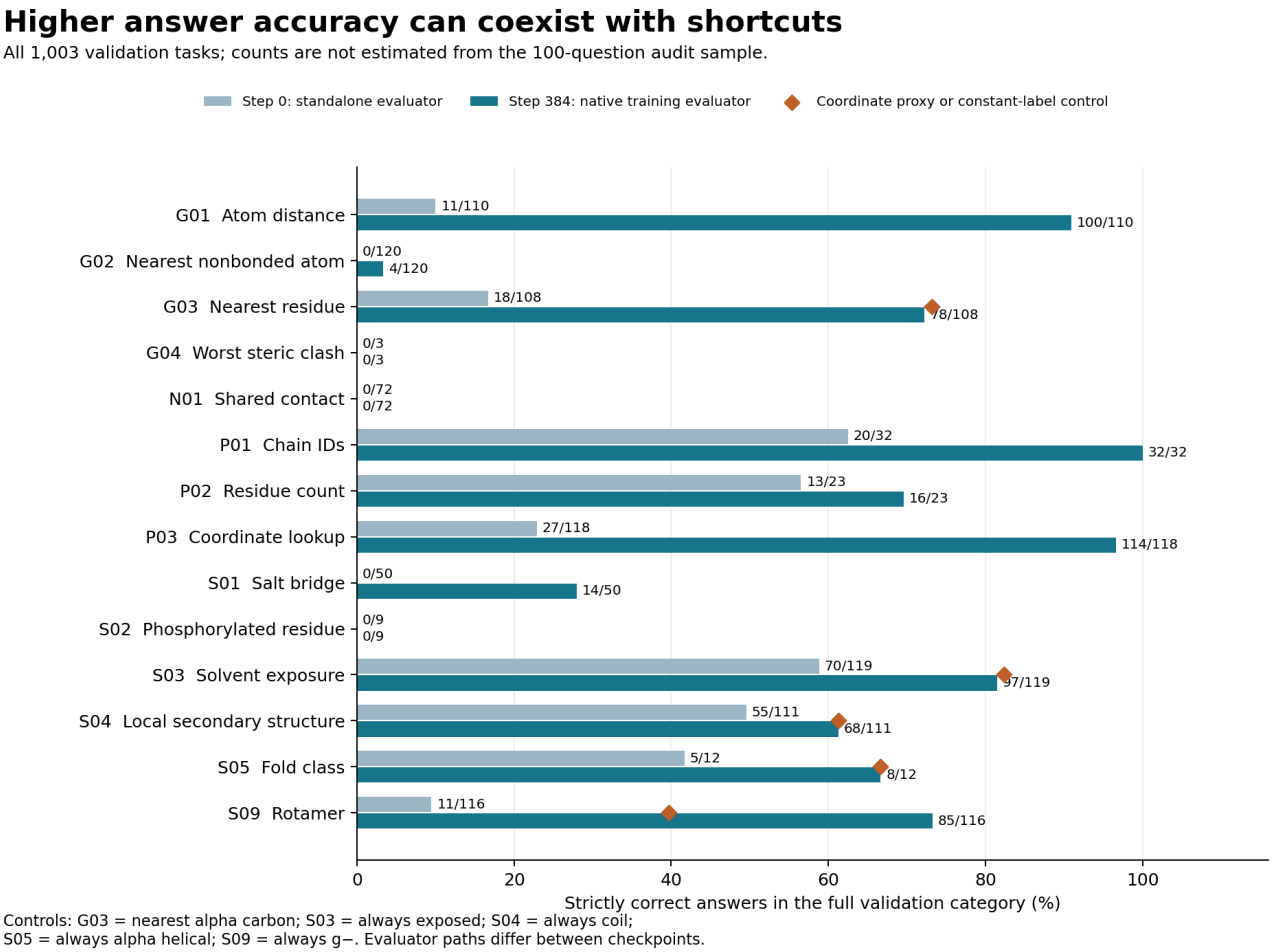
All 200 responses were inspected through text views that remove exact duplicate lines, show beginning/middle/end sections of long responses, and support targeted examination of computational passages. Repeated trailing invented metadata is retained in the raw artifact; the normal review view focuses on the substantive answer before the first JSON metadata block. Truncation and parsing failures were checked separately, including post-answer repetition. This was a qualitative Codex review, not a blinded independent-human or multiple-rater study. **Not every arithmetic line of every long trace was independently recomputed.**

Independent numerical checks parse fixed-column coordinates directly from the displayed prompts. They recompute all 116 S09 gold dihedrals using two equivalent formulas, all 108 G03 heavy-atom minima and their CA-only controls, all seven sampled G01 distances, all seven sampled P03 coordinate answers, and the true charged-atom contact for all seven sampled S01 questions. All recomputed gold answers agree. Unit checks cover known signed torsions, proper rotation/translation/scale invariance, a bin-changing counterexample to the wrong formula, and signed coordinate parsing.

The review distinguishes **correct final label**, **valid observable method**, **shortcut**, **format repair** and **unfinished generation**. A model’s prose is evidence of what it wrote, not proof of the internal mechanism that caused its answer. Claims about computation refer to visible operations; mentions of DSSP, Python or a database are not evidence that a tool ran. No tools were available to these model rollouts.

Family	Full N	Sample 00 / 01 / 10 / 11	Family	Full N	Sample 00 / 01 / 10 / 11
G01	110	2 / 2 / 0 / 3	P03	118	2 / 3 / 0 / 2
G02	120	3 / 4 / 0 / 0	S01	50	4 / 3 / 0 / 0
G03	108	2 / 2 / 1 / 2	S02	9	7 / 0 / 0 / 0
G04	3	3 / 0 / 0 / 0	S03	119	2 / 1 / 2 / 2
N01	72	7 / 0 / 0 / 0	S04	111	2 / 2 / 2 / 1
P01	32	0 / 4 / 0 / 3	S05	12	3 / 2 / 0 / 2
P02	23	2 / 2 / 2 / 1	S09	116	3 / 3 / 3 / 4

Full-cohort scores and simple controls



Controls: G03 = nearest alpha carbon; S03 = always exposed; S04 = always coil;
S05 = always alpha helical; S09 = always g−. Evaluator paths differ between checkpoints.

Bars: full-category strict accuracy at each archived checkpoint, with numerator/denominator labels. Diamonds: explicitly defined offline controls. These are conditional accuracies on context-eligible validation questions; the sample is used for qualitative interpretation only. The different inference paths prevent treating every bar difference as an isolated RL effect.

The controls require no learned molecular-reasoning policy: G03 selects the nearest CA; S03 always says exposed; S04 always says coil; S05 always says alpha helical; S09 always says g−. The S09 constant is clearly worse than the model. The S03/S04/S05 controls equal or exceed the model, so their score gains do not establish structural computation.

Category findings: P01 and P02

P01 · Chain identifiers

Full validation: **20/32 → 32/32**. Formatting and completion improve; the sampled chain-identification knowledge was already present.

Both checkpoints identify the right chain in all seven sampled questions. The four scored gains are output-contract repairs: boxed/inline FINAL is not parsed, or a trailing “Final Answer” overrides G with ANSWER. Step 384 gives a clean final field. These examples are single-chain structures.

How it fails: No failure remains at step 384 in the 32-question validation category. That does not establish more complex multichain parsing, nor does this sample show new molecular geometry knowledge.

P02 · Residue counting

Full validation: **13/23 → 16/23**. More consistent enumeration in some cases, with substantial counting and interpretation errors.

Case 41 recovers the 59-residue count after the baseline omits two positions. Case 42 enumerates 60 residues correctly where the baseline lists60 but claims83. Both checkpoints can count the 38-residue example in case 43.

How it fails: Case 39 regresses from 78 to 150 by mistaking the last author residue number for the count. Case 38 counts 22 then wrongly excludes TPO as a nonprotein triphosphate. Case 40 invents an extra entry. Case 44 gives the correct 29 but continues generating a repeated sequence until the length cap, so native reward is zero.

Category findings: P03 and G01

P03 · Coordinate lookup

Full validation: **27/118 → 114/118**. Strong evidence of improved retrieval and faithful copying of supplied coordinates.

The three sampled gains repair wrong coordinate content, including incorrect signs/components and wrong residue lookup. Case 49 changes x from 5.668 to the supplied−6.841; case 50 finds the requested R349 NH1 rather than another coordinate triple. The full category improves from 27/118 to 114/118.

How it fails: Lookup is still imperfect. Case 51 returns the SE atom of MSE1 instead of MSE33 at both checkpoints. Case 47 locates the correct residue context but substitutes a wrong CG2 triple. These tasks test reading the PDB, not deriving a physical quantity.

G01 · Atom distance

Full validation: **11/110 → 100/110**. Stronger atom lookup plus actual Euclidean distance calculation.

The model uses coordinate differences, squares, sums and a square root. This is substantive task-relevant behavior, already present in some baseline answers and substantially more reliable at step 384. In case 1, the baseline uses D73/L79 instead of D80/L94; step 384 selects the requested atoms and obtains23.86 Å. Case 3 now finishes the computation instead of exhausting its output budget.

How it fails: Residual failures include wrong residue/atom lookup, arithmetic and formatting. Case 6 substitutes a different oxygen and invents a serine CG. Case 2 correctly computes12.20 Å but never supplies a parseable FINAL field. Case 3 gives21.72 for a gold21.708 and passes the 0.02 Å tolerance despite a square-root error.

Category findings: G02 and G03

G02 · Nearest nonbonded atom

Full validation: **0/120 → 4/120**. Some coordinate-based search and bond-exclusion reasoning; no robust general solution.

All four step 384 successes in the entire120-question category are included because the success stratum is so small. They compute distances to alpha carbons and selected sidechain atoms, then sometimes recover the correct allowed atom in the target residue. Cases 10–13 show progress from sequence guesses, missing-target claims or unfinished loops.

How it fails: The searches are incomplete, and the covalent-bond graph is unreliable. Case 8 selects directly bonded GLU8 CB even though it must be excluded. Case 9 uses an incorrect arginine connectivity and selects CB instead of CG. Three-letter or otherwise invalid atom identifiers are common. Four successes out of 120 is weak evidence of general nearest-nonbonded-atom competence.

G03 · Nearest residue

Full validation: **18/108 → 78/108**. A clear coordinate shortcut: nearest alpha carbon substitutes for minimum heavy-atom distance.

All seven sampled step 384 traces explicitly compare CA–CA distances. Their answers all match the independently computed CA proxy. On the full108-question category,104 predictions match that proxy. The proxy itself scores 79/108; the model scores 78/108. This strongly supports a useful but incomplete geometric heuristic.

How it fails: The proxy is wrong on29 questions. The model follows its wrong answer on27 of those 29, and solves only one. In case 15, C192 is closer by CA distance (5.010 vs6.253 Å), but D191 is closer by the required heavy-atom minimum (2.976 vs3.298 Å). Cases 16 and18 show the same structural failure, with additional coordinate-copying mistakes.

Category findings: G04 and N01

G04 · Worst steric clash

Full validation: **0/3 → 0/3**. No demonstrated success at finding the worst steric clash.

The entire validation category has only three questions, and all three are reviewed. Step 384 makes more numerical attempts, including distance and radius arithmetic, but no response computes the required maximum positive atom-pair overlap reliably.

How it fails: It substitutes CA distances or residue-average radii, assumes unmeasured contact distances, mishandles bonded exclusions and omits relevant nonpolymer atoms. Cases 21 and23 report the least-negative overlap despite the requirement for a positive clash. Case 22 assumes2.8 Å contacts and averages residue radii. All three final identifiers are also invalid.

N01 · Shared contact residue

Full validation: **0/72 → 0/72**. More arithmetic, but the wrong contact definition and no demonstrated solution.

Baseline answers largely speculate about sequence midpoints or get stuck listing residues. All seven step 384 samples attempt CA-distance screens at the 4 Å cutoff, whereas the task requires any heavy-atom pair within 4 Å to each anchor.

How it fails: All seven trained responses truncate without a final answer; full-category performance remains0/72. A CA pair farther than 4 Å cannot rule out a sidechain contact. Misnumbered residues and repeated rescanning compound this invalid screening rule. Numerical effort alone is not evidence that the contact-intersection task was learned.

Category findings: S01 and S02

S01 · Salt-bridge partner

Full validation: **0/50 → 14/50**. Some real charged-atom distance calculations, especially for lysine partners.

Cases 54,56 and58 compute the specified oxygen–nitrogen distances and return the correct partners. Independent checks recover E860 OE2–K889 NZ=3.174 Å, E9 OE2–K13 NZ=2.863 Å and D6 OD1–K3 NZ=3.210 Å. Case 55 computes the correct R170–E162 contact but uses the wrong residue identifier format.

How it fails: Arginine searches often stop at NE or use CA/carboxylate-carbon proxies, missing NH1/NH2. Cases 52,53 and57 all have a qualifying NH1 contact even though every NE-to-partner-oxygen distance exceeds4 Å. Those step 384 traces truncate. This explains a concrete limit of the apparent salt-bridge procedure.

S02 · Phosphorylated residue

Full validation: **0/9 → 0/9**. No scored improvement, and several sampled answers lose correct biochemical identification.

The baseline often recognizes SEP as phosphoserine or TPO as phosphothreonine but fails to convert to the requested parent one-letter residue identifier, e.g. Y:S26. Step 384 still recognizes TPO in cases 62 and64, but omits the residue number.

How it fails: In cases 60,63 and65, step 384 instead calls SEP/TPO a separate small molecule, sulfate or nucleotide and guesses an ordinary Ser/Thr. Case 59 ignores phosphate atoms and guesses a common serine site. All seven sampled responses and all nine validation tasks fail. This category illustrates why a zero-to-zero score can hide semantic regressions.

Category findings: S03 and S04

S03 · Solvent exposure

Full validation: **70/119 → 97/119**. A near-constant solvent-exposed answer; no demonstrated solvent-accessibility calculation.

All seven sampled step 384 answers assume a typical accessible area or invent a relative SASA value, without computing occlusion from surrounding atoms. Across all 119 tasks,118 parsed answers are solvent-exposed and one is unparsed; none is buried.

How it fails: Always answering solvent-exposed scores 98/119, slightly above the model's97/119. All 21 buried questions fail. Cases 68 and69 regress from correct burial heuristics to exposure. The score increase is therefore not evidence that the model learned to calculate solvent accessibility.

S04 · Local secondary structure

Full validation: **55/111 → 68/111**. A near-constant coil answer; no demonstrated DSSP or backbone-geometry computation.

All seven sampled step 384 answers say coil. They refer to imagined DSSP runs, generic sequence propensities or invented phi/psi values. No actual tool is available or called. Across111 tasks,110 parsed predictions are coil and one is unparsed.

How it fails: Always answering coil scores 68/111, exactly the model's score. It solves none of the 33 helix or10 strand questions. Cases 73 and74 lose correct helix guesses. The apparent gain reflects class selection and output reliability, not demonstrated secondary-structure assignment.

Category findings: S05 and S09

S05 · Overall fold class

Full validation: **5/12 → 8/12**. An alpha-helical class preference and sequence-propensity rhetoric, without structural calculation.

The sampled responses do not calculate hydrogen bonds, backbone angles or helix/strand fractions. At step 384,10 of 12 full-category predictions are predominantly alpha helical and two are mixed. All eight correct answers are alpha-helical chains.

How it fails: Always answering alpha helical also scores 8/12. The single beta-sheet and all three mixed chains fail. Cases 81 and85 show wrong helix guesses; case 80 remains wrong mixed. One sampled gain (case 82) merely formats the baseline’s already correct helix guess.

S09 · Sidechain rotamer

Full validation: **11/116 → 85/116**. A more consistent explicit torsion-calculation template, but not a validated dihedral algorithm.

All 13 sampled step 384 responses attempt coordinate subtraction, cross products and angle calculation. Nine start with the correct four coordinates; four have clear extraction errors. Seven receive reward1, but none survives all checks of coordinates, formula, arithmetic and bin mapping. This is more than an always-g− rule, yet less than reliable rotamer computation.

How it fails: A recurring formula omits the middle-bond norm; there are also wrong atom selections, sign transcription errors, lost atan2 quadrants and positive-angle-to-g− bin mistakes. Correct category rewards frequently tolerate an incorrect angle. The dedicated rotamer section shows exactly why.

Rotamers: calculation or trick?

The answer is a mixture: **the visible step-384 behavior is an explicit calculation template, not simply a constant rotamer guess, but it is not a reliable dihedral algorithm.** It extracts N, CA, CB and a residue-specific fourth atom (e.g. CG or OG1), forms bond vectors and cross products, estimates an angle, and assigns one of three bins. The full category has 46 g−, 37 trans and 33 g+ labels; step 384 scores 85/116 versus 46/116 for always g−.

Let A=N, B=CA, C=CB and D=the fourth atom. Define u=B−A, v=C−B, w=D−C, n_0=u×v and n_1=v×w. With the benchmark's signed convention, a correct formula is:

χ_1 = atan2(‖v‖ (n_0 · w), n_0 · n_1), converted to degrees and wrapped to [−180°,180°).

A recurring trace formula is **atan2(n_0·w, n_0·n_1)**, missing the factor ‖v‖. The numerator and denominator then have different length dimensions, and changing coordinate scale changes the computed angle. This is not fixed by merely using atan2. Other traces subsequently replace terms, lose the quadrant, or make arithmetic errors.

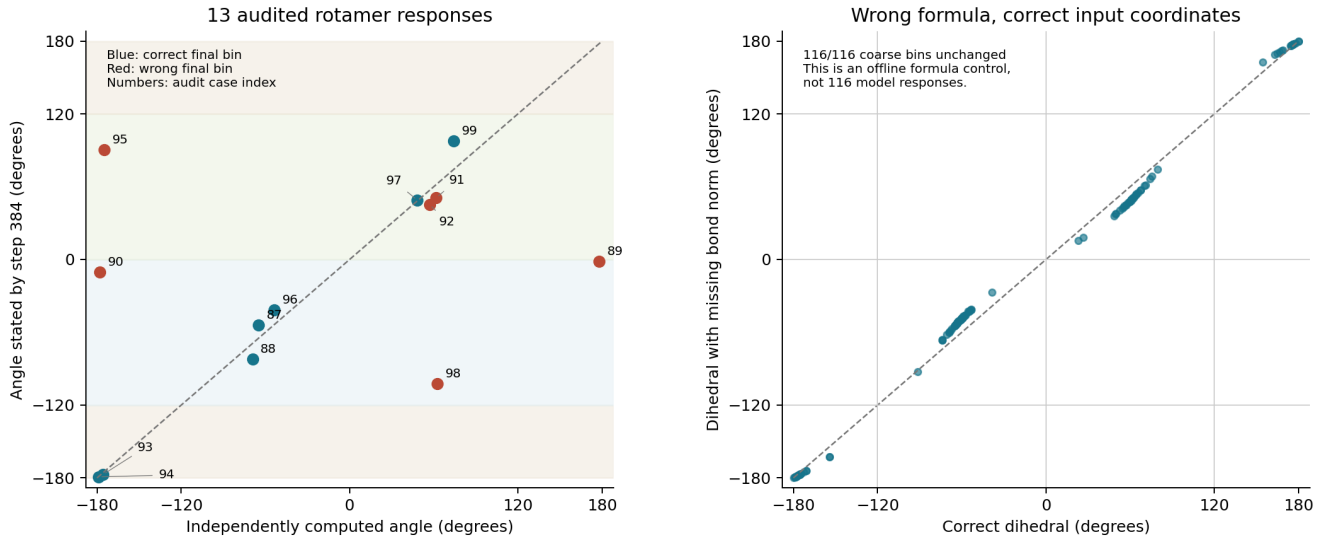
Yet applying this wrong formula to the correct four coordinates for every S09 validation question preserves **116/116 bins**. That is an offline formula control, not a claim that the model calculated all 116 angles correctly. The benchmark asks for broad bins: g− is −120°≤χ<0°, g+ is0°≤χ<120°, and trans is the rest. The observed geometries do not expose this formula error at those boundaries. A synthetic115° dihedral with a1.53-length middle bond crosses from g+ to trans under the wrong formula, so the shortcut is not generally valid.

In the 13 reviewed step 384 traces, four have clearly wrong input coordinates, two map a positive computed angle to g−, and one explicitly loses the atan2 quadrant. These counts are review findings in a biased sample, with overlapping error types. None of the 13 passes all stages of an end-to-end computation; nevertheless seven final bins are rewarded. At step 0, some correct bins likewise follow invented coordinates or a guess. Correctness of the baseline label is not a trustworthy procedural control either.

This is consistent with reward favoring a reusable, often successful template that does not need exact intermediate mathematics. It does not establish deliberate exploitation or prove that the written rationale causes the final answer. Stronger causal evidence would require controlled interventions on coordinates and comparison under matched inference settings.

Rotamer diagnostic plots

Rotamer reward does not certify the angle calculation



Bins: g− = [−120°, 0°); g+ = [0°, 120°); t = remaining angles. Correct bins can follow incorrect mathematics.
Left: outcome-stratified sample, not a population accuracy estimate; stated angles include transcription and arithmetic errors.

Left: independently recomputed angle versus the last numerical angle discussed in the sampled step384 trace; annotations are zero-based audit case indices. Blue means correct final bin, red wrong final bin. Angles are periodic, and the numbers reflect the model's own inconsistent calculations, not a clean implemented formula. Right: the isolated missing-norm formula on correct input coordinates for all116 questions. This separates a formula defect from input-copying and later arithmetic defects.

Case	True angle / bin	Step384 statement	Why the score is misleading or fails
87	−64.851° / g−	about−54° / g−	Right coordinates; missing norm; correct category.
89	+177.692° / t	−1.508° / g−	atan(y/x) used in place of atan2; quadrant lost.
90	−177.982° / t	about−10.3° / g−	Lys NZ substituted for backbone N.
91	+61.758° / g+	about+50.8° / g−	Positive bin mislabeled g− after coordinate arithmetic.
92	+57.067° / g+	about+45.14° / g−	Positive-angle sign/bin mistake, despite correct input points.
94	−178.782° / t	about−179.2° / t	Missing norm, but correct quadrant/wrapping and category.

Implications for the RL result

What is supported: the policy became much better at reading requested coordinates and completing simple distance calculations in this validation distribution. Some salt-bridge responses also use the right charged-atom geometry. Those are meaningful improvements. The separate matched original-benchmark result is still encouraging, although its uncertainty and limited coverage remain as reported.

What is not supported: a broad claim that the model learned reliable molecular structure analysis. Exposure and local secondary-structure gains are explained by constant labels. Fold classification reaches only the majority-class score. Nearest-residue success mostly follows a CA-only approximation. Rotamer categories often pass while the intermediate geometry is wrong. The hardest searches remain mostly or entirely unsolved.

These findings explain why aggregate validation accuracy is an incomplete progress measure, especially after the context filter changes class frequencies. They qualify the step 384 peak. They do **not** by themselves identify the cause of the later384→512 optimization regression, which requires the separate training/runtime audit in the main report.

A useful next experiment would keep the existing benchmark frozen and add a separate diagnostic set: balanced buried/exposed and helix/strand/coil labels; nearest-residue cases where CA and heavy-atom rankings disagree; rotamers near±120° and0°; atom-name distractors such as N versus NZ; and rigidly rotated copies plus chemically valid torsion changes. A continuous-angle diagnostic would reveal errors hidden by coarse bins. No such new model experiment is claimed in this report.

Selection and calibration matter: step 384 was chosen using validation, this sample is conditioned on outcomes, several questions share a structure, and inference paths differ. A stronger learning claim should reproduce the behavior on fresh source groups under matched serving, with repeated decoding where appropriate. The current audit provides concrete hypotheses and failure examples for that work, not an independent test-set estimate.

Reproducibility files: sample-manifest.json (seed, allocation, selected IDs and hashes), sample.json.gz (full prompts and200 raw responses), annotations.json and review-notes.json (all100 case reviews), statistics.json (full-category counts), geometry.json (independent checks), and rollout-audit.html (filterable paired viewer). Code: sample_reasoning_audit.py, analyze_reasoning_audit.py, plot_reasoning_audit.py and build_reasoning_audit.py. All paths are relative to this experiment; artifacts are in results/reasoning-audit/.

Appendix: every reviewed pair

Indices are stable, zero-based positions in the frozen sample. Outcome states use the native strict scores. Notes describe visible reasoning, including cases where the correct answer content is not accepted by the scorer. Full text and prompt are available under the same index in the HTML viewer.

Case 0 · G01 · state 11

pdbthink-g01-37da72df37f7fdc06014
Gold: 15.035

Step 0: Correct atom lookup and Euclidean method; mistaken square self-correction still within tolerance.

Step 384: Correct lookup and Euclidean calculation; final 15.04 agrees within tolerance.

Case 1 · G01 · state 01

pdbthink-g01-42d1e9b9cd0c41faec01
Gold: 23.86

Step 0: Uses D73 OD1 and L79 CB instead of requested D80/L94; computes 20.58 and malformed final.

Step 384: Uses requested atoms and computes 23.86 correctly.

Case 2 · G01 · state 00

pdbthink-g01-8c9718bbadeec9826f24
Gold: 12.2

Step 0: Wrong coordinates; computes 19.08 and boxed final not parsed.

Step 384: Correct coordinates and 12.20 distance, but says "We'll present FINAL" inline; no parsed FINAL.

Case 3 · G01 · state 01

pdbthink-g01-9b4bd938bcad18869011
Gold: 21.708

Step 0: Copies long atom lists and truncates before finishing.

Step 384: Correct atoms and squared sum; root arithmetic slightly wrong (21.72), accepted within 0.02 tolerance of 21.708.

Case 4 · G01 · state 11

pdbthink-g01-c90fc18d48008cadf422
Gold: 19.54

Step 0: Correct coordinate lookup and distance 19.54 after correcting a square.

Step 384: Correct coordinate lookup and distance 19.54; long repeated checking.

Case 5 · G01 · state 11

pdbthink-g01-d32a21518d4be1f2a28a
Gold: 7.363

Step 0: Correct atoms and Euclidean method; small arithmetic error, 7.36 accepted.

Step 384: Correct atoms, squared sum and distance 7.36.

Case 6 · G01 · state 00

pdbthink-g01-e410c580c153c1b78514
Gold: 15.703

Step 0: Wrong NZ/OG coordinates; computes 18.04 rather than 15.703.

Step 384: Correct NZ, initially sees correct OG but substitutes a different residue's coordinates and invents a serine CG; computes 15.07.

Case 7 · G02 · state 00

pdbthink-g02-0a13ea30c05bcb6f5882
Gold: "B:R347:CB"

Step 0: Sequence-neighbor speculation, excludes same residue too broadly; malformed B:ARG348:CA.

Step 384: Partial distance search, confuses sequence/residue names and selects nonexistent generic VAL CG; malformed answer.

Case 8 · G02 · state 00

pdbthink-g02-45fa8c56214f3ef68d5a
Gold: "E:P9:CD"

Step 0: Checks distances and correctly identifies PRO9 CD, but writes three-letter E:PRO9:CD.

Step 384: Computes many distances but selects directly bonded GLU8 CB, which must be excluded; three-letter identifier also fails.

Case 9 · G02 · state 00

pdbthink-g02-93369b0b6d0088581bfa
Gold: "B:R52:CG"

Step 0: Uses neighboring CA distances as proxy for target CZ; guesses PRO53 CA; malformed final.

Step 384: Wrong arginine connectivity excludes CG incorrectly; selects CB instead of CG, also malformed identifier.

Case 10 · G02 · state 01

pdbthink-g02-a05334aaddb277a3ac2e
Gold: "A:L7:CA"

Step 0: Incorrect leucine connectivity; guesses from sequence and truncates.

Step 384: Computes target-to-CA distances, correctly keeps own CA three bonds away; correct answer but no exhaustive all-atom proof.

Case 11 · G02 · state 01

pdbthink-g02-a7b76ffd50d590819814
Gold: "B:R349:CD"

Step 0: Repeated claims calculation impossible without tools; truncates.

Step 384: CA screening plus selected sidechain distances identifies own CD at about 2.828 A; correct answer, incomplete global search.

Case 12 · G02 · state 01

pdbthink-g02-e13b8e044eeff159245b
Gold: "A:D140:CA"

Step 0: Long listing, speculative sequence neighbors, confused exclusions, truncation.

Step 384: Target-to-CA screening and own-residue checks; excludes CB and picks CA at 3.629 A; correct answer without exhaustive atom search.

Case 13 · G02 · state 01

pdbthink-g02-eac05bcbd6cf3cac559d
Gold: "A:R893:CB"

Step 0: Enumerates atoms, falsely claims target missing, loops to truncation.

Step 384: Computes many CA and selected sidechain distances; correctly identifies own CB about 2.909 A, checks three-bond eligibility.

Case 14 · G03 · state 11

pdbthink-g03-aaa38eebbd31bc7e1b20
Gold: "A:V97"

Step 0: Chooses sequence-nearest V97, explicitly assumes chain linear; right label for invalid rationale.

Step 384: Uses CA-CA distances, V97 wins; correct label but not requested heavy-atom minimum.

Case 15 · G03 · state 00

pdbthink-g03-b8c52dcf813b5fdbaf68
Gold: "A:D191"

Step 0: Sequence-nearest C192 guess, boxed unparsed.

Step 384: Explicitly substitutes CA-CA distance and chooses C192; true all-heavy-atom winner D191.

Case 16 · G03 · state 00

pdbthink-g03-b8dd716f48aaca153aa8
Gold: "A:K254"

Step 0: Guesses K254 using sequence/sidechain arguments, even suggests Lys-Lys salt bridge; malformed final.

Step 384: Uses CA-CA proxy with miscopied coordinates; picks G255, also the true CA-proxy winner, instead of all-heavy-atom winner K254.

Case 17 · G03 · state 01

pdbthink-g03-b8e42ce58c960e7a9581
Gold: "A:V128"

Step 0: Sequence-nearest V128 guess is right but boxed unparsed.

Step 384: Numerical CA-CA comparison picks V128; correct label via proxy.

Case 18 · G03 · state 10

pdbthink-g03-d39788371b897d2c6854
Gold: "A:A726"

Step 0: Confused lookup and sequence guess picks A726 correctly.

Step 384: Uses CA-CA proxy (one coordinate is miscopied) and picks R727; true CA-proxy winner is also R727, but all-heavy-atom winner is A726.

Case 19 · G03 · state 11

pdbthink-g03-d3d17304af054d3a8090
Gold: "C:S15"

Step 0: Attempts selected atom distances and picks S15 correctly.

Step 384: Only CA-CA distances, picks S15 correctly.

Case 20 · G03 · state 01

pdbthink-g03-ff62f34e606e9930f42c
Gold: "B:R9"

Step 0: Misidentifies ASN as LYS and gets stuck listing atoms; truncates.

Step 384: Recognizes ASN and computes CA-CA distances, picks R9 correctly.

Case 21 · G04 · state 00

pdbthink-g04-55369d6376f773133850
Gold: "E:E8--E:P9"

Step 0: Attempts many distances but repeats identical estimates and truncates.

Step 384: Partial pair search finds no positive overlap, then reports least-negative TPO--MET; invalid criterion and identifier.

Case 22 · G04 · state 00

pdbthink-g04-ec33a251c9d7f1fddc18
Gold: "C:L122--C:T21"

Step 0: CA-distance shortcut with miscopied coordinates; picks TPO14--SER15, invalid identifier.

Step 384: Uses residue-average radii and assumed 2.8 A separation instead of closest atom pair; guesses LEU--GLU, invalid identifier.

Case 23 · G04 · state 00

pdbthink-g04-ed3c5d17f53366383bb7
Gold: "A:G1--A:L10"

Step 0: Excludes nonpolymer ligand, loops about needing tools and truncates.

Step 384: CA-CA surrogate and selected contacts; chooses least-negative overlap GLU9--ILE10, misses ligand clash and invalid identifier.

Case 24 · N01 · state 00

pdbthink-n01-042badfe4ccddfb8399e
Gold: "A:S33"

Step 0: Confused residue numbering and midpoint/sequence guesses, loops to truncation.

Step 384: Enumerates CA distances with 4 A cutoff instead of all-heavy-atom contacts; truncates without answer.

Case 25 · N01 · state 00

pdbthink-n01-0e296986440388362055
Gold: "A:D729"

Step 0: Midpoint heuristic guesses residue729 correctly but omits amino-acid code.

Step 384: CA-distance screen finds no shared contact and repeats calculations to truncation.

Case 26 · N01 · state 00

pdbthink-n01-6aba22c1ed9650543bde
Gold: "A:N34"

Step 0: Confused sequence/residue identity, midpoint speculation, truncation.

Step 384: CA-distance screen and even self-contact speculation, truncates without finding N34.

Case 27 · N01 · state 00

pdbthink-n01-6c054aa0e97241324843
Gold: "B:E66"

Step 0: Sequence-based guess GLU69, wrong partner.

Step 384: Misnumbers residues, computes CA distances, then repetitive verbal loop; truncation.

Case 28 · N01 · state 00

pdbthink-n01-9c4ded25b7879e22b8a0
Gold: "A:F696"

Step 0: Misidentifies CYS as MSE; midpoint guess696, missing residue code and refusal flag.

Step 384: CA-distance screening with confused identities, truncates without finding F696.

Case 29 · N01 · state 00

pdbthink-n01-dde84c4aa13319672701
Gold: "A:L257"

Step 0: Sequence-position/lysine prior guesses K258, boxed unparsed.

Step 384: Scans CA distances rather than atom contacts; truncates without finding L257.

Case 30 · N01 · state 00

pdbthink-n01-f3ee21661ab54b5f2428
Gold: "A:K5"

Step 0: Invents unbounded A:A residue list to thousands, truncates.

Step 384: Numerical CA-distance screening but wrong contact surrogate; truncates.

Case 31 · P01 · state 11

pdbthink-p01-1c5de69b97a988c5ff37
Gold: ["A"]

Step 0: Correctly reads single chain A.

Step 384: Correctly reads single chain A, concise FINAL.

Case 32 · P01 · state 11

pdbthink-p01-3626b2c2edd92266bd9e
Gold: ["B"]

Step 0: Correctly reads single chain B.

Step 384: Correctly reads single chain B, concise FINAL.

Case 33 · P01 · state 01

pdbthink-p01-4e6d164114d60bc16ef6
Gold: ["Y"]

Step 0: Correctly reads chain Y, but FINAL only inline/boxed and unparsed.

Step 384: Same correct chain Y in valid FINAL format.

Case 34 · P01 · state 01

pdbthink-p01-886b952a14d72529f9e7
Gold: ["E"]

Step 0: Correctly reads chain E, but FINAL only inline/boxed and unparsed.

Step 384: Same correct chain E in valid FINAL format.

Case 35 · P01 · state 01

pdbthink-p01-b04a7082daaf0d04c2d7
Gold: ["G"]

Step 0: Correct chain G, but trailing "Final Answer" overrides parsed value with ANSWER.

Step 384: Correct chain G with clean final field.

Case 36 · P01 · state 11

pdbthink-p01-da81c82017f235e96bd8
Gold: ["A"]

Step 0: Correctly reads single chain A.

Step 384: Correctly reads single chain A, concise FINAL.

Case 37 · P01 · state 01

pdbthink-p01-f1b3aaf8b39d4ea2727d
Gold: ["C"]

Step 0: Correctly reads chain C, but FINAL only inline/boxed and unparsed.

Step 384: Same correct chain C in valid FINAL format.

Case 38 · P02 · state 00

pdbthink-p02-0e516812e3e00d5bb293
Gold: 22

Step 0: Lists22 residues but miscounts21.

Step 384: Counts 22 then wrongly excludes TPO as non-amino-acid triphosphate; answers21.

Case 39 · P02 · state 10

pdbthink-p02-3ba49f7068c2ac316262
Gold: 78

Step 0: Correct inclusive span150-73+1=78.

Step 384: Enumerates using author numbers73..150 then confuses last author number with count; answers150.

Case 40 · P02 · state 00

pdbthink-p02-3fc51a2a8e831be302f1
Gold: 35

Step 0: Lists35 distinct numbers, correct 35 but malformed final.

Step 384: Invents/duplicates sequence entries and residue368; answers36.

Case 41 · P02 · state 01

pdbthink-p02-535fc17ac77638245254
Gold: 59

Step 0: Omits residues12 and32 and returns57.

Step 384: Enumerates 59 positions and answers59; some sequence transcription remains inconsistent.

Case 42 · P02 · state 01

pdbthink-p02-556576bc66cde9224992
Gold: 60

Step 0: Enumerates 60 entries yet claims83; first residue also wrong.

Step 384: Correctly enumerates 224..283 as60.

Case 43 · P02 · state 11

pdbthink-p02-6837b5c90341cbb2aeed
Gold: 38

Step 0: Correctly enumerates 38 residues.

Step 384: Correctly enumerates 38 residues.

Case 44 · P02 · state 10

pdbthink-p02-dbad3107f376709840b0
Gold: 29

Step 0: Correct 29, excludes ligand.

Step 384: Correct 29 and valid FINAL, but generation continues and hits length limit; native truncation makes reward0. The raw tail is a long repeated sequence string after the correct answer.

Case 45 · P03 · state 11

pdbthink-p03-0deb94b2b5395dea3aff
Gold: [-4.689, 1.288, 6.099]

Step 0: Finds requested atom and copies correct coordinates.

Step 384: Finds requested atom and copies correct coordinates.

Case 46 · P03 · state 11

pdbthink-p03-1005fd0d9a9a7e260de1
Gold: [2.366, 1.446, 1.744]

Step 0: Finds requested atom and copies correct coordinates.

Step 384: Finds requested atom and copies correct coordinates.

Case 47 · P03 · state 00

pdbthink-p03-78deef03b65a3ee5debc
Gold: [1.185, 2.082, 3.636]

Step 0: Reports wrong coordinate triple for I189 CG2.

Step 384: Correct residue context but substitutes another coordinate triple for CG2; wrong answer.

Case 48 · P03 · state 01

pdbthink-p03-d6e1a628b1a14a793f0f
Gold: [0.524, -12.107, -3.51]

Step 0: Correct x but wrong y/z for Q32 CD.

Step 384: Correct full Q32 CD triple.

Case 49 · P03 · state 01

pdbthink-p03-dcf450aa37957ad9e600
Gold: [-6.841, -6.857, 3.708]

Step 0: Wrong x (5.668 rather than -6.841) for I724 CD1.

Step 384: Correct full I724 CD1 triple.

Case 50 · P03 · state 01

pdbthink-p03-e325d678f44432b07f87
Gold: [-7.831, -21.44, -1.959]

Step 0: Wrong NH1 coordinates and malformed final.

Step 384: Correct R349 NH1 coordinates and format.

Case 51 · P03 · state 00

pdbthink-p03-f4f1255c38a7abc2b4f3
Gold: [3.394, 4.922, -1.942]

Step 0: Returns SE of MSE1 rather than MSE33.

Step 384: Same MSE1-for-MSE33 error, now correctly formatted.

Case 52 · S01 · state 00

pdbthink-s01-14cf6f33b6243fb40ff2
Gold: "A:D151"

Step 0: Wrong/mixed coordinates; concludes GLU125 and encodes glutamate as G.

Step 384: Attempts acidic partners but CA proxies and incomplete nitrogen search; never finishes. Independent check: NH1-OD2=3.764 A satisfies contact, whereas both NE-OD distances exceed4 A.

Case 53 · S01 · state 00

pdbthink-s01-1898ea7f0fb5738b95c3
Gold: "A:D725"

Step 0: Sequence-based acidic-partner guess ASP685; malformed final.

Step 384: Calculates CA and then NE distances, confused acidic identities, truncates before complete nitrogen search. Independent check: NH1-OD1=3.768 A satisfies contact, whereas both NE-OD distances exceed6 A.

Case 54 · S01 · state 01

pdbthink-s01-387af46b559a66878f81
Gold: "A:K889"

Step 0: Sequence-nearest basic residue guess K876; malformed identifier.

Step 384: Uses explicit E860 OE1/OE2 to K889 NZ distances3.799/3.174 A, correct partner.

Case 55 · S01 · state 00

pdbthink-s01-a2888039258cd7aa6a8f
Gold: "A:E162"

Step 0: Sequence-neighbor speculation picks GLU162, but malformed identifier.

Step 384: Computes NE-to-OE1/OE2 distances3.745/2.696 A and finds correct partner, but writes A:Glu162 rather than A:E162.

Case 56 · S01 · state 01

pdbthink-s01-b0f0c59d7ce76313a097
Gold: "A:K13"

Step 0: Sequence/helix heuristic guesses Lys13 correctly, malformed identifier.

Step 384: Checks charged-atom distances and finds Lys13 NZ within 4 A of Glu9 OE2, correct one-letter identifier.

Case 57 · S01 · state 00

pdbthink-s01-f33d19b7091e744c8119
Gold: "A:D127"

Step 0: Atom/residue-number confusion and endless listing, truncates.

Step 384: Mostly uses Arg NE and carboxylate carbon/oxygen proxies, misidentifies residues and truncates. Independent check: NH1-OD1=2.999 A satisfies contact; NE-OD distances are4.375/4.756 A.

Case 58 · S01 · state 01

pdbthink-s01-fa708c8afeed9655d2c8
Gold: "B:K3"

Step 0: Starts coordinate lookup then unbounded invented sequence repetition, truncates.

Step 384: Checks Lys/Arg nitrogen to Asp6 oxygen distances and identifies Lys3; corrects chainA toB before final.

Case 59 · S02 · state 00

pdbthink-s02-0e901babb294a47b1aed
Gold: "X:S5"

Step 0: Incorrectly calls SEP selenocysteine, loops to truncation.

Step 384: Rejects SEP as nonprotein/triphosphate despite phosphate atoms, picks generic Ser by frequency prior; invalid identifier.

Case 60 · S02 · state 00

pdbthink-s02-4944f8404e8565535336
Gold: "B:S1493"

Step 0: Correctly identifies SEP1493 as phosphoserine but omits number in final.

Step 384: Misidentifies SEP as sodium ethyl phosphate and chooses terminal SER; invalid identifier.

Case 61 · S02 · state 00

pdbthink-s02-733ea89e853501b52c10
Gold: "B:S1212"

Step 0: Rejects SEP as phosphorylatable, speculates about other S/T/Y residues, truncates.

Step 384: Selects ordinary Ser1184/sequence-position14 from priors; invalid S:14 identifier.

Case 62 · S02 · state 00

pdbthink-s02-7f11b16bfe86ea3f49f3
Gold: "E:T5"

Step 0: Correctly identifies TPO5 as phosphothreonine but uses three-letter modified code.

Step 384: Eventually identifies TPO but omits number and required T one-letter code.

Case 63 · S02 · state 00

pdbthink-s02-b793e33d783844f71cb6
Gold: "Y:S26"

Step 0: Correctly identifies SEP26 as phosphoserine, but uses SEP instead of S.

Step 384: Wrongly calls SEP sulfate/nonprotein and infers Thr25 phosphorylation; invalid identifier.

Case 64 · S02 · state 00

pdbthink-s02-e98afd677b93afdd2b37
Gold: "E:T5"

Step 0: Correct TPO5 identification, wrong modified residue code.

Step 384: Recognizes TPO5 phosphothreonine but outputs E:TPO without number.

Case 65 · S02 · state 00

pdbthink-s02-f6e1880304ac9eb7dec6
Gold: "G:T378"

Step 0: Correctly identifies TPO378 but malformed identifier.

Step 384: Wrongly calls TPO phosphorylated thymidine/nucleotide, picks ordinary SER385 from frequency prior.

Case 66 · S03 · state 00

pdbthink-s03-1a87d8bc274e61c8c679
Gold: "buried"

Step 0: Default solvent-exposed guess, wrong buried Arg.

Step 384: Invents typical Arg RSA around0.45, no coordinate occlusion computation; wrong solvent-exposed.

Case 67 · S03 · state 01

pdbthink-s03-1eade67d68f9e20a8837
Gold: "solvent-exposed"

Step 0: Falsely claims P58 absent and loops/refuses to truncation.

Step 384: Invents typical Pro RSA0.45/0.46, no actual surface calculation; correct exposed label.

Case 68 · S03 · state 10

pdbthink-s03-375152b0490a9f864d2d
Gold: "buried"

Step 0: Misidentifies Glu as Asp; interior-position heuristic gives correct buried label.

Step 384: Uses charged-residue prior and invented typical RSA0.52; wrong exposed label.

Case 69 · S03 · state 10

pdbthink-s03-66571901f19bad228378
Gold: "buried"

Step 0: Hydrophobic Ile burial heuristic gives correct label.

Step 384: Assumes typical Ile environment is exposed, no computed SASA; wrong label.

Case 70 · S03 · state 11

pdbthink-s03-76c290cc180df4c0bb0e
Gold: "solvent-exposed"

Step 0: Polar/linear-chain heuristic gives correct exposed label.

Step 384: Invented typical GLN RSA0.45-0.55, no actual geometry; correct exposed label.

Case 71 · S03 · state 00

pdbthink-s03-93d1d33b8b0f9b3d7612
Gold: "buried"

Step 0: Gly/N-terminal exposure heuristic, wrong label and malformed final.

Step 384: Assumes Gly SASA near maximum, gives wrong exposed label.

Case 72 · S03 · state 11

pdbthink-s03-aee73415e190766108d4
Gold: "solvent-exposed"

Step 0: C-terminal Pro exposure guess gives correct label.

Step 384: Invented typical Pro SASA values, mentions Shrake-Rupley without executing it; correct exposed label.

Case 73 · S04 · state 10

pdbthink-s04-064ef4ff0d977acbdbba
Gold: "helix"

Step 0: Sequence helix-propensity guess happens to be correct.

Step 384: Claims simulated DSSP gives coil without calculations; wrong label.

Case 74 · S04 · state 10

pdbthink-s04-1113ec03b05888fda20c
Gold: "helix"

Step 0: Leucine helix-propensity guess happens to be correct.

Step 384: Misidentifies Leu145 as Glu and invents typical phi/psi values; wrong coil.

Case 75 · S04 · state 01

pdbthink-s04-4c7551ab5335ff00be01
Gold: "coil"

Step 0: Coil guess based on absent annotations, boxed unparsed.

Step 384: Conceptual Python comment asserts coil, no DSSP/angle calculation; right label and format.

Case 76 · S04 · state 01

pdbthink-s04-80cb05fdaeadb52db361
Gold: "coil"

Step 0: Leucine helix guess, malformed final.

Step 384: Repeated simulated-DSSP claims and coil prior, right label.

Case 77 · S04 · state 00

pdbthink-s04-ceb6ea12c11f39a9614a
Gold: "helix"

Step 0: Default coil in absence of annotations, wrong helix.

Step 384: Claims typical DSSP/known result coil, no actual calculation; wrong helix.

Case 78 · S04 · state 11

pdbthink-s04-d19843e976ccbfbe45c2
Gold: "coil"

Step 0: Defaults coil despite confusion over whether target exists; correct label.

Step 384: Invented typical phi/psi and inconsistent structural ranges, final coil correct.

Case 79 · S04 · state 00

pdbthink-s04-d973760836233043d5af
Gold: "helix"

Step 0: Polar-residue/loop propensity guess coil, wrong helix.

Step 384: Claims absent 3D data and imagined DSSP result coil, wrong helix.

Case 80 · S05 · state 00

pdbthink-s05-1da4447909b792b66e68
Gold: "predominantly alpha helical"

Step 0: Defaults mixed alpha/beta from chain length/lack of annotations; wrong helix.

Step 384: Imagined DSSP and generic composition prior, chooses mixed despite briefly saying helix; wrong.

Case 81 · S05 · state 00

pdbthink-s05-2b02ff803542cc679d69
Gold: "predominantly beta sheet"

Step 0: Default mixed in absence of annotations, wrong beta.

Step 384: Helix propensity/imagined DSSP claims, wrong beta.

Case 82 · S05 · state 01

pdbthink-s05-6491a9f0ba5f8398cf12
Gold: "predominantly alpha helical"

Step 0: Sequence helix-propensity guess correct, boxed unparsed.

Step 384: Same general helix-prior reasoning, no geometry, correct final format.

Case 83 · S05 · state 11

pdbthink-s05-935fd4833f2d5d88ac61
Gold: "predominantly alpha helical"

Step 0: Helix assumption without calculated angles or H bonds; correct label.

Step 384: Long confused atom/residue enumeration then generic helix prior; correct label.

Case 84 · S05 · state 11

pdbthink-s05-9922e3fbacc94f0967f1
Gold: "predominantly alpha helical"

Step 0: Invents long sequence span and helix rationale, correct label.

Step 384: Imagined DSSP and composition prior, no calculated structure, correct helix.

Case 85 · S05 · state 00

pdbthink-s05-ba17ccb48c5c29215a24
Gold: "mixed alpha/beta"

Step 0: Hesitant mixed guess is semantically correct but refusal/unparsed.

Step 384: Generic helix frequency prior changes answer to wrong helix.

Case 86 · S05 · state 01

pdbthink-s05-f90567232a33288834f5
Gold: "predominantly alpha helical"

Step 0: Unbounded repeated invented sequence, truncation.

Step 384: General helix-propensity argument, no geometry, correct helix and completion.

Case 87 · S09 · state 01

pdbthink-s09-00e61ebcad77f449d5ac
Gold: "g-"

Step 0: Correct input points but wrong torsion denominator (normal dot middle bond, zero by construction); -90 degrees gives right g-, boxed FINAL unparsed.

Step 384: Correct four points; atan2 omits middle-bond norm; -54 rather than -64.851 degrees, still g-.

Case 88 · S09 · state 01

pdbthink-s09-05ad620dea98eba272f9
Gold: "g-"

Step 0: Invented input coordinates and wrong norm-product denominator; -16.6 degrees and g- unparsed.

Step 384: N x sign copied incorrectly; missing-norm formula and inaccurate atan estimate yield about -82 degrees; g- correct despite wrong geometry.

Case 89 · S09 · state 10

pdbthink-s09-1238597964105a6e0234
Gold: "t"

Step 0: Invented coordinates and chemical misidentification; approximately -178.55 degrees yields correct t.

Step 384: Correct points; omits norm and replaces atan2 with atan(y/x), losing quadrant: -1.508 degrees instead of +177.692; wrong g-.

Case 90 · S09 · state 10

pdbthink-s09-18bb0a0277996b7c14dc
Gold: "t"

Step 0: Confuses NZ with N, later invents a different coordinate set; 169.5 degrees yields correct t.

Step 384: Uses terminal lysine NZ instead of backbone N; missing-norm calculation gives -10.3 degrees and wrong g- (true -177.982/t).

Case 91 · S09 · state 00

pdbthink-s09-27908dbdd70df9caa959
Gold: "g+"

Step 0: Misidentifies residue and later uses wrong coordinates; -34.6 degrees and malformed g-.

Step 384: Initially correct points, later CB-z transcription error; about +50.8 degrees but labels positive interval g-; true +61.758/g+.

Case 92 · S09 · state 00

pdbthink-s09-65f1805f38c56084285b
Gold: "g+"

Step 0: Unbounded atom enumeration, no finished angle or FINAL.

Step 384: Correct points; missing-norm angle +45.14 degrees, falsely claims prompt duplicates g- bin, answers g-; true +57.067/g+.

Case 93 · S09 · state 11

pdbthink-s09-6ea578199e0816dc8e23
Gold: "t"

Step 0: Invented coordinate set, inconsistent formulas; -170.9 degrees yields correct t.

Step 384: Correct points; missing-norm -177.326 degrees versus true -175.917; handles quadrant and t correctly.

Case 94 · S09 · state 01

pdbthink-s09-714945209b1ba67d620c
Gold: "t"

Step 0: Invented coordinate set and wrong formula; 103.5 degrees/g+, scoring also marks refusal.

Step 384: Correct points; missing-norm -179.2 degrees versus true -178.782; correct wrapping and t.

Case 95 · S09 · state 10

pdbthink-s09-8100dc4b872225eab41e
Gold: "t"

Step 0: Invented points include implausibly long bond; wrong formula still gives -177 degrees and correct t.

Step 384: Wrong N coordinates, other three correct; missing-norm 90.6 degrees/g+ instead of true -174.724/t.

Case 96 · S09 · state 11

pdbthink-s09-b432ff6735b8a390c2d6
Gold: "g-"

Step 0: Correct four coordinates but cross-product component and sign errors; eventually chooses correct g-.

Step 384: Correct points; missing-norm -41.6 degrees instead of -53.650, correct g-.

Case 97 · S09 · state 11

pdbthink-s09-c2f5987840cc6e243a68
Gold: "g+"

Step 0: Wrong N/CA coordinates and wrong norm-product formula; 38.3 degrees gives correct g+.

Step 384: Correct points but missing norm plus cross/dot/arctangent arithmetic errors; switches from 56.5 to 48.9 degrees, right g+ (true 48.318).

Case 98 · S09 · state 00

pdbthink-s09-df09cf8da6916725f250
Gold: "g+"

Step 0: Hallucinates coincident N/CA, repeats undefined-torsion argument and truncates.

Step 384: Drops x signs and mixes/fabricates N/CA components; -102.5 degrees/g- instead of true +62.575/g+.

Case 99 · S09 · state 11

pdbthink-s09-e07a5c69968b5c356336
Gold: "g+"

Step 0: Explicit educated rotamer-prior guess with invented coordinates; no completed valid calculation, correct g+.

Step 384: Correct initial points; starts missing-norm formula, then substitutes unrelated cross-product component into denominator; 97.5 degrees gives correct g+ (true 74.291).